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The Editors
Computational Statistics & Data Analysis

Dear Editors,

Hereby, I submit the manuscript "CausalMixGPD: An R Package for Bayesian Nonparametric Conditional Density Modeling in Causal Inference and Clustering with a Heavy-Tail Extension" for potential publication in Computational Statistics & Data Analysis. This is a computational statistics software contribution in line with the scope of the journal.

Applications often involve heavy tails because the outcomes are rare risks/costs/policies. Parametric models do a good job capturing the central portion but miss the tail whereas flexible nonparametric techniques fail at capturing the extremes. I introduce CausalMixGPD, an R package based on Dirichlet process mixtures for the central part and generalized Pareto tail using extreme value theory. To the best of our knowledge, no such package exists which implements a distributional interface with predictor-dependent clustering and causal inference.

Unconditional and covariate-dependent spliced models can be fitted with posterior inference using nimble, with posterior predictive distributions of density, survival, quantiles, mean, and restricted-mean functional values (including extreme quantiles). Predictor-dependent Dirichlet process mixture clustering and causal inference for binary treatment and continuous outcome can be carried out, where I provide estimates for six estimands (ATE, ATT, QTE, QTT, CATE, CQTE) with propensity-score adjustment for observational data. All the above methods share a unified predictive framework and are implemented via R generics. Examples of clustering and causal inference, such as Lalonde (1978) earnings data are provided as reproducible illustrations.

The software package CausalMixGPD v0.8.0 is GPL-3 licensed and is available at CRAN; development version is available at https://github.com/arnabaich96/CausalMixGPD, where all manuscript results are reproducible.

I certify the manuscript is original, unpublished and not currently being considered by other journals. No conflicts of interest and funding sources to declare.

Sincerely,

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